

1.6 Aluminium Evaporation Source

Electrically-heated evaporators are located within an enclosure called the source box. The source box is made from sections of aluminium, and is internally water-cooled.

At approximately 750 amps and 10 volts, the evaporators reach an operating temperature of 1500°C.

Aluminium wire is delivered to each of the evaporator from stepper motor-driven wire feeders. In the vacuum environment, the aluminium condenses on the substrate as it passes the source.

This machine has a single source fixed within the chamber. It is designed to allow easy access for cleaning and maintenance.

The inline control system monitors the uniformity of aluminium deposition across the web. It automatically sends signals to the stepper motors, changing the amount of aluminium wire to achieve uniform deposition across the substrate.

Viewing windows in the chamber wall allow the operator to monitor the evaporators and make any necessary adjustments.



Figure 1.6-1 Aluminium evaporation coating source area

1.7 Machine Control Systems

The control systems allow the users to operate the machine.

Machine Components

The machine control system gathers information from:

- computer and programmable logic controllers, or PLCs
- communication links
- monitoring devices

The machine modifies parameters using information about:

- coating measurements
- power levels to the evaporators
- vacuum levels
- wire feed rates
- plasma process gas flow rates
- drive system parameters
- chiller system operational feedback data.

The computer uses the profibus and profinet protocols to transmit information between system components.

At the end of a coating cycle, the machine can provide a data report. [See chapter 9.13 Reporting Options/Data Logging on page 485 for details.](#)

Human-Machine Interface (HMI)

The operator interacts with the machine using a human-machine interface, or HMI. The HMI can be used by touching the screen, or attaching a keyboard and mouse.

When the main isolator is turned on, the Log On screen is displayed on the HMI.

To secure the HMI, access to some screens is limited by passwords.



Figure 1.7-1 Operator HMI interface screen

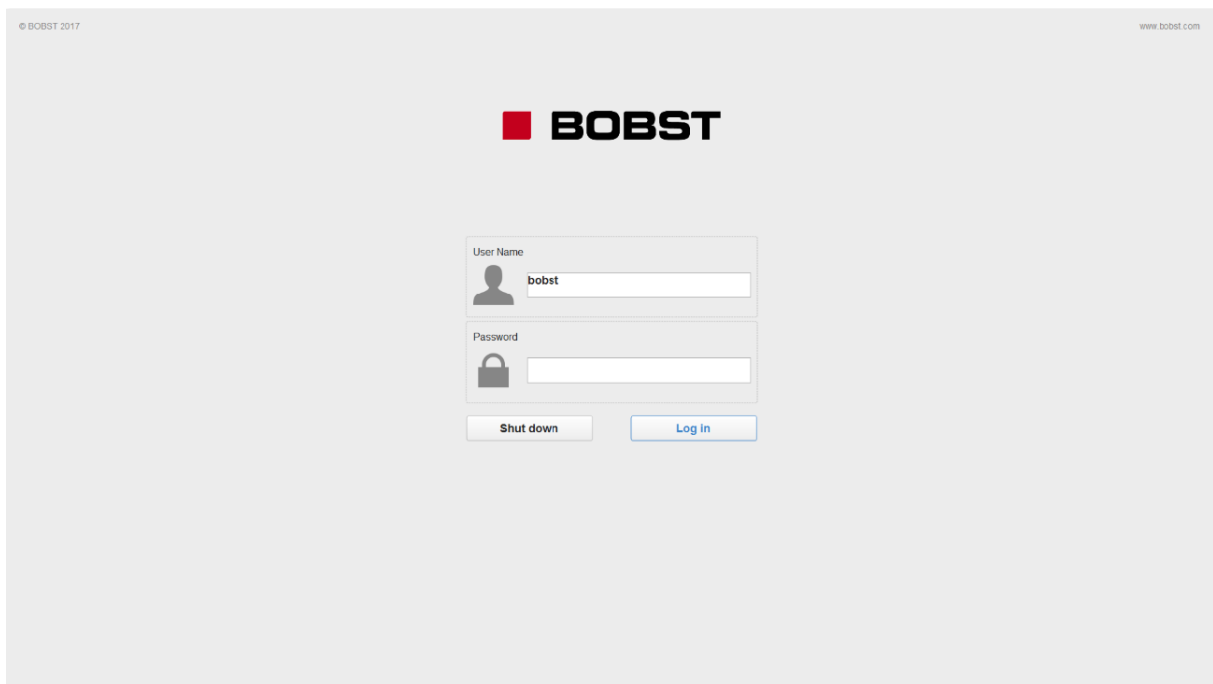


Figure 1.7-2 HMI Log On screen

SECTION 2.0 VACUUM PUMPING SYSTEM

This section describes the vacuum pumping system on the K5 Expert in the following chapters:

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2.1 Vacuum Pumping System Requirements

A vacuum environment has to be created within the chamber to allow the coating process to take place. The vacuum reduces the density of the gas molecules in the chamber to minimise contamination of the deposition .

A number of different types of vacuum pumps are used to achieve the process conditions required. The selection of these pumps is made to allow an efficient evacuation of the gases from the chamber to a level where the process can take place.

The vacuum chamber is divided into two vacuum zones by various sealing methods within the chamber:

- The evaporation area (zone) is required to be at a level less than 8×10^{-4} mbar
- The winding zone is required to be in a level of approximately, 2×10^{-2} mbar.

The vacuum measurement system being used is mbar, with atmosphere being 1000 mbar. Gauging instrumentation is provided to give the user feedback on the vacuum levels, in different areas of the machine.

The design of the vacuum system gives benefits including lower power consumption due to smaller pumping group requirements, lower pressure levels in the evaporation zone and improved cleanliness of the winding zone.

A combination of the different types of pumps are used to evacuate the chamber and provide the vacuum level required.

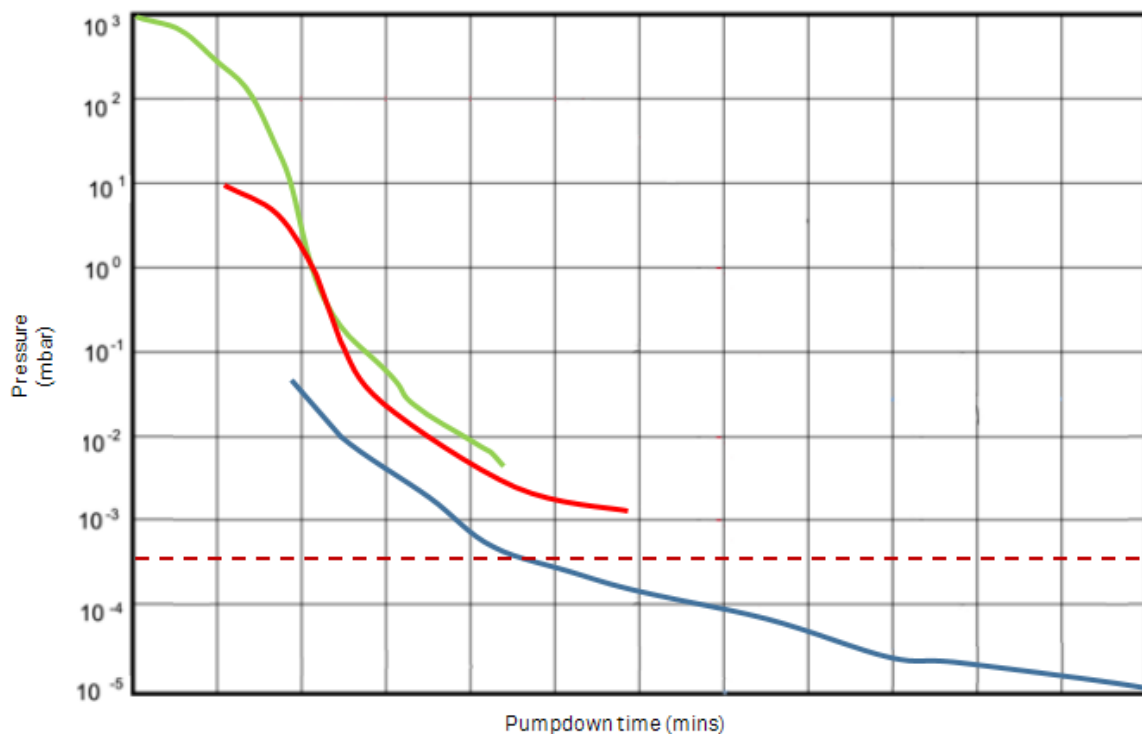


Figure 2.1-1 Curve indicating combination of vacuum pumps

Key:-

- Busch 630 Dry pump pumping
- AERZEN Booster pump pumping
- Varian Diffusion pump pumping

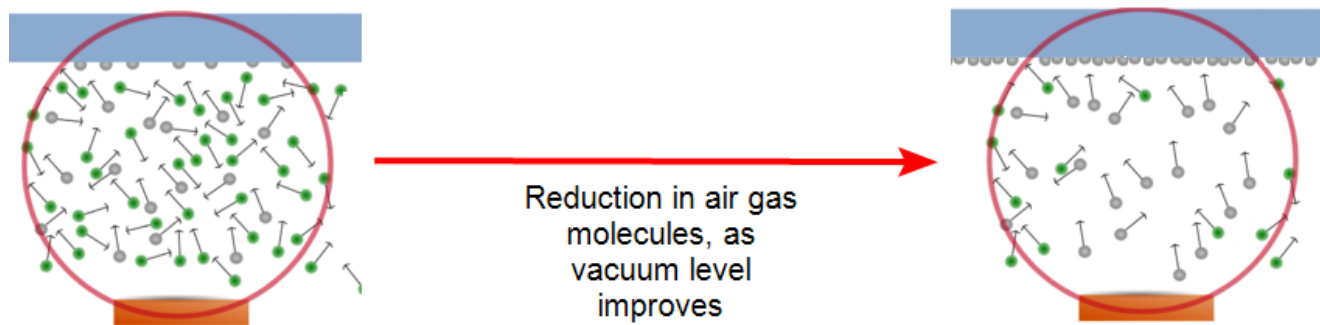


Figure 2.1-2 Display of effect of vacuum level on evaporation process

2.2 Vacuum Pumping System Mimic Display Screen

The vacuum pumping system is controlled automatically in a pre-defined sequence to evacuate the chamber safely, while protecting its system components.

To allow the user to understand what is occurring with the machines vacuum system, a HMI screen is provided.

Figure 2.2-1 Vacuum HMI screen

Select the SETTINGS and then DIAGNOSTICS button.

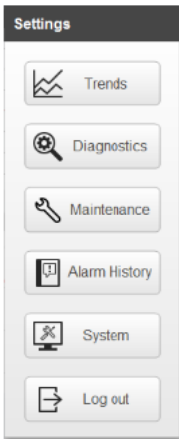


Figure 2.2-2 Example of the settings menu

Select the VACUUM tab from the top of the DIAGNOSTICS HMI screen, and the VACUUM HMI screen will be displayed.

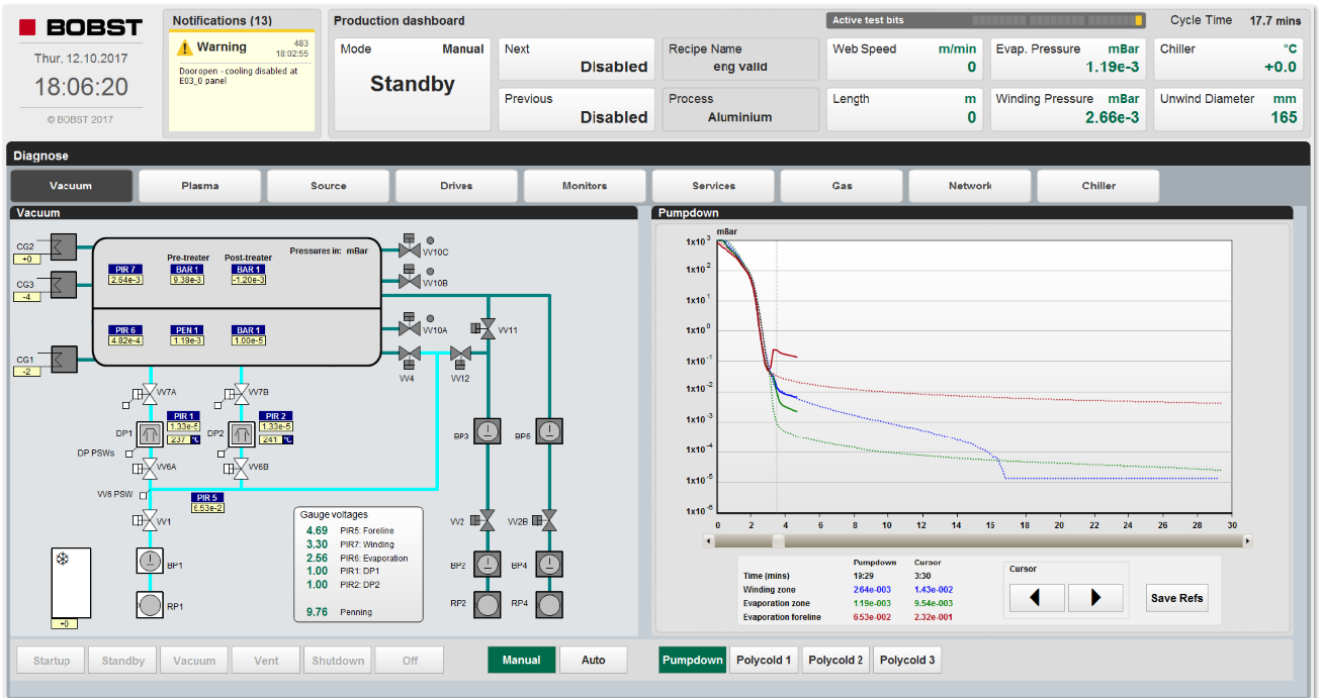


Figure 2.2-3 Example of the diagnose vacuum menu

The screen has a number of buttons, on the right-hand side, that control the operating mode of the vacuum pumping system.

Button	Mode of Operation
Vacuum	Places the vacuum pumping system into a pumping condition to evacuate the chamber

Button	Mode of Operation
Standby	Places the vacuum pumping system in a standby condition
Vent	Vents the chamber to atmosphere
Startup	Starts all the required pumps from a non-energised condition
Shutdown	Places the vacuum pumping system in a controlled shutdown sequence
Auto	When highlighted, the system is in the AUTO mode
Manual	When highlighted, the system is in the MANUAL mode

Table 2.2-1 Vacuum HMI screen operating buttons

The pumps and valves are colour coded to show their operating condition;

- Grey - Closed or Off
- White - Open or Running
- Alternating Grey and White - Opening, but not fully open, or Closing, but not fully closed
- Red - Neither open or closed, but in a 'fault' condition

In the maintenance access level, manual control of the vacuum system components can be made by selecting the manual button

A touch screen facility has been supplied to allow the user the facility to start individual pumps and open valves safely when in this "manual" mode.

2.3 Vacuum Pumping System Control

As previously mentioned, the vacuum pumps and valves are controlled from the “VACUUM” HMI interface screen.

Full protection and interlocking of the pumping-group are provided by the machine PLC system, using sensors on the machine. The machine PLC system is used together with interfaces on the valves and pressure monitors located at critical parts of the system.

Normal operation of the machine is in the operator mode. If the machine control has been operating in the supervisory mode for set up or maintenance reasons, ensure that it is placed back in the normal operating mode.

The AUTO and MANUAL buttons can be used in the maintenance mode to change the normal operation between the auto and manual modes.

The button is coloured green to Indicate which mode is selected.



Figure 2.3-1 Auto and manual selection buttons on the VACUUM HMI screen

Manual Operation – Maintenance Access Level

The vacuum isolation valves and pumps can be operated directly from the VACUUM screen. This facility is useful during testing and maintenance procedures.

	Note!
	Note that all the safety interlocks remain in operation so that the vacuum pumping system will not allow any unsafe operation, e.g. the chamber cannot be evacuated unless the winding cart is fully engaged within the chamber.

Automatic Operation – Operator Access Level

When switched to 'Automatic', control of individual valves and pumps are not permitted. This is the normal operating mode of the vacuum pumping system.

In the 'Automatic' mode, operation of the 'vacuum' button starts the sequence of starting the pumps and opening the isolation valves. This allows the quickest possible evacuation of the chamber to coating conditions, under safe conditions.

Operational modes

The vacuum pumping system has five operating modes

Startup

From a cold/previously powered down condition, the vacuum pumping system can be brought on line using this start command, if all interlocks are met.

If the system will not start when the button is pressed, a pop up message box will be shown giving the reason for the “non-action”

	Note!
	Once the diffusion pumps are at operating temperature, this mode will be deselected and the control mode will change to “HOLD” condition, and will be indicated to the user

Vacuum Mode

When this mode of operation is selected, the air contained within the chamber will be evacuated by the pumping system. The level of vacuum reached will be generally lower than 8×10^{-4} mbar in the evaporation zone.

Interlocks are used to ensure that this operation only occurs in a safe manner.

If the machine is already in this mode of operation the vacuum buttons are disabled.

Standby Mode

This mode is selected to place the pumping system in a standby condition. The pressure level in the chamber will generally not change, when this mode is selected.

With the machine in the non-coating condition, the control of the pumping system has been designed to allow the pumping system to stay in a state of readiness, but at the same time using the minimum amount of power.

Vent Mode

When this mode is selected, the chiller and polycolds are reheated. When the source cools, the valves (vent) are energized to allow the air surrounding the chamber back into the chamber. The pressure levels inside and outside the chamber will equalise.

Interlocks are used to ensure that this operation only occurs in a safe manner.

Shutdown mode

Selecting this mode of operation will shutdown the vacuum pumping system sequentially.

It normally takes between 30-50 minutes for this action to take place and is dependent upon the cooling water supplied to the machine.

Eco Vacuum mode

Eco vacuum is an energy saving mode of operation. It is provided to power down certain elements of the vacuum pumping system, when the chamber is opened.

This control mode is selected from the CONFIGURATION – MODES 2 HMI screen

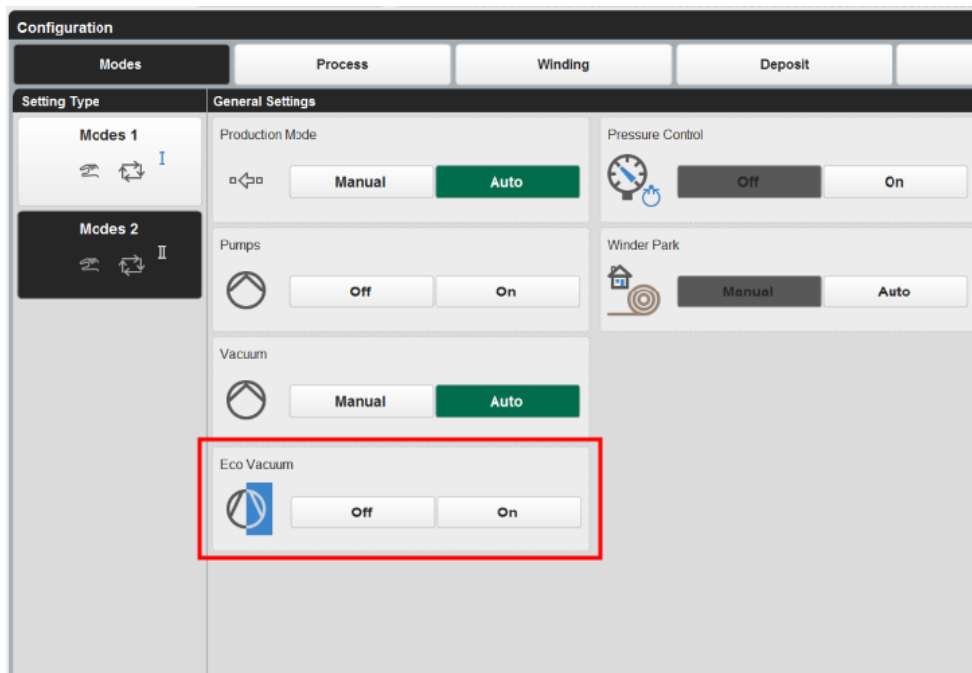


Figure 2.3-2 CONFIGURATION MODES HMI screen

Diffusion Pump - Temperature Sensors

The diffusion pumps installed on the machine have PT100 temperature sensors to monitor the internal oil temperature.

These temperature values are displayed to the user on the VACUUM HMI screen.

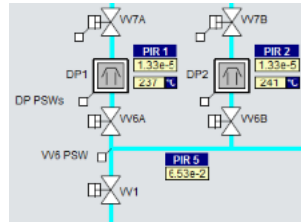


Figure 2.3-3 Vacuum HMI showing the diffusion pump temperature feedback

	Note!
	When the diffusion pumps are not at the operating temperature, the vacuum sequence cannot be started. The normal temperature should be 220 C.

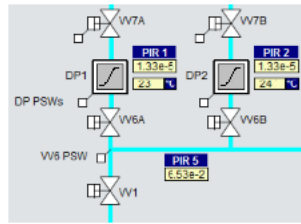


Figure 2.3-4 Vacuum HMI showing the diffusion pump indicating "warming up" condition

Brookes Automation Cryogenic Unit Control

The cryogenic pumping unit, is interlocked and controlled by the machine.

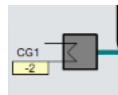


Figure 2.3-5 Example Vacuum HMI screen showing cryogenic unit icon

The icon will change colour dependent upon the operation mode that it has been switched to while the vacuum system is in the "automatic" mode

- White – Standby Mode
- Light Blue – Cool Mode
- Yellow – Defrost

2.4 Vacuum Pumping System Start-Up Sequence

This chapter describes the vacuum pumping system start up sequence.

Water and Air Supply Requirements

The vacuum pumping system requires compressed air supply at a minimum pressure of 5.5 bar (approximately 80 psi) and cooling water supply at a minimum pressure of 4 bar (approximately 60 psi) at the various cooling water manifolds, to operate.

The condition of these interlocks can be checked from the HMI diagnostics – Services screen.

When these requirements are met the water and air indicators will be white.

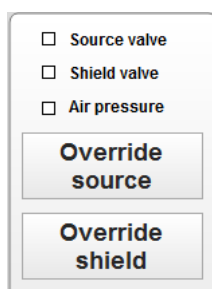


Figure 2.4-1 HMI Diagnostic screen - Services

	Caution !
	The process drum chiller unit should be energised for a minimum of four hours before the unit is placed in the cool operating mode.

Start-Up of the Vacuum Pumping System

Press the Startup button to start the process chiller unit, diffusion pump heaters, cryogenerator units and the rotary pumps when the pumping control system is in the Automatic mode of operation. This is accessible from the SETTINGS-DIAGNOSTICS-VACUUM selection buttons.

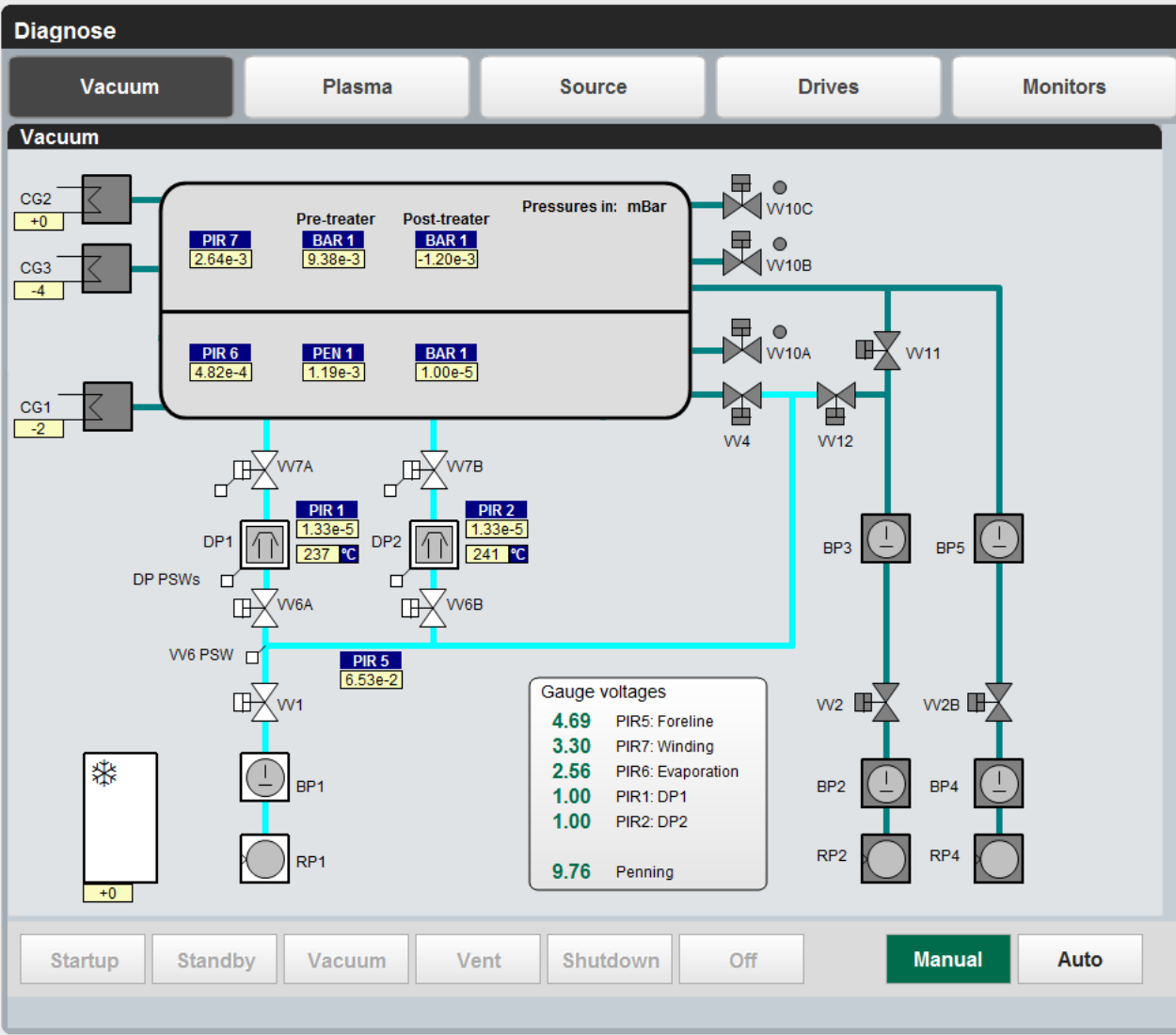


Figure 2.4-2 VACUUM HMI diagnostic screen

2.5 Types of Pumps Incorporated on Machines

The vacuum environment is created by a number of suction (vacuum) pumps that remove air from the vacuum chamber and discharges it to the surrounding atmosphere.

Several different types of pump are used to efficiently transition from atmospheric pressure to 10^{-5} mbar:

- Rotary screw pumps have a comparative low pumping speed, but will pump a high-pressure differential between input and output, for example 1 mbar inlet to 1000 mbar outlet.
- A mechanical booster pump (Roots type blower) has a high pumping speed, but will only work against a small differential pressure between inlet and outlet ports. The pressure range of this type of pump is 10 to 10^{-3} mbar
- At very low pressures, e.g., below 10^{-2} mbar, diffusion pumps are used to provide a very high speed of pumping.
- In vacuum coaters, where high volumes of water vapour are present, a cryogenic pumping system is used to condense and freeze the gases to coils within the chamber.

Rotary Screw Pumps

Principle of operation:

Two parallel screws rotate in opposite directions in the pump body. Entering gases are trapped between the pitches of the screws and the pump body. The gases are conveyed by the rotation of the screws to the exhaust side where they are discharged.

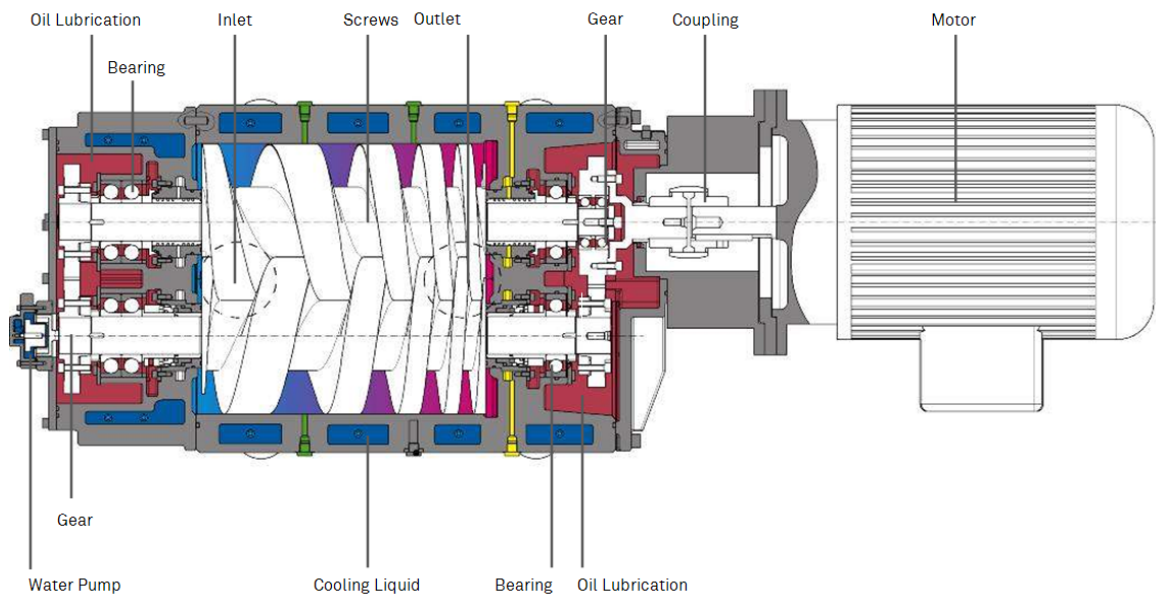


Figure 2.5-1 Dry Screw Pump

Mechanical Booster Pump

A typical mechanical booster pump utilizes two figures of eight shaped impellers, synchronized by external gears and rotating in opposite directions inside the stator. The gas is trapped between the impellers and the stator wall and transferred from the high vacuum inlet to the discharge side of the pump. The gears are oil lubricated and are isolated from the impellers, so that the impellers run dry. The clearance between the impellers and the stator is generally about 0.005 to 0.025 cm. Therefore, back-leakage of gas occurs at a rate governed by the pressure difference between input and output and type of gas being pumped. Under normal running conditions, a pressure difference of about ten to one is obtained across the pump.

Pressure Range 10 – 10^{-3} mbar

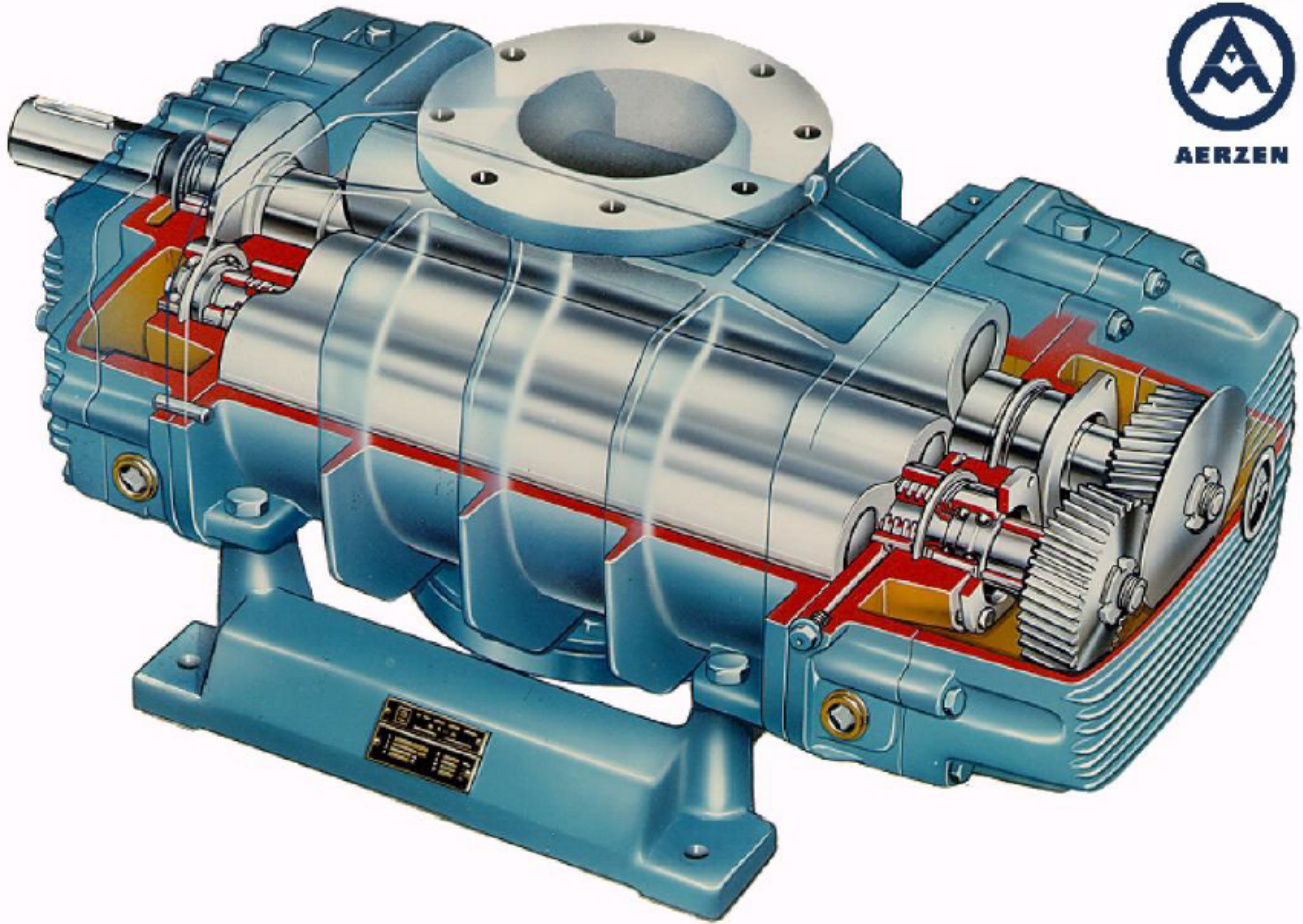


Figure 2.5-2 Cut away drawing of mechanical booster pump

This type of pump can be also described as a "Roots" or "Blower" type pump.

Depending upon the size and type of machine, the numbers of this type of pump will vary

This type of pump is used as a second roughing stage to bring the vacuum level rapidly down to a level of 5×10^{-2} mbar before the high vacuum valves can be opened.

They consist of two impellers that rotate inside the pump body. The action of the movement is to sweep gases from the inlet side to the outlet side of the pump.

Variable Frequency Drive (VFD)

All booster pumps installed on the machine have their speed controlled by panel mounted variable frequency drives, located in the electrical panel (E03_0)

This principle of operation allows the pumps to be used earlier in the pumpdown cycle without overloading them.

Diffusion Pumps

The high vacuum valves open when the roughing pumps have brought the vacuum level inside the chamber to a level below 7×10^{-2} mbar. This allows the diffusion pumps to pump the evaporation zone, while the roughing pumps are pumping the winding zone.

The level of vacuum that this type of pump will achieve will depend upon the condition of the evaporation zone and the material being coated. This range is normally 1×10^{-3} mbar down to 8×10^{-5} mbar during metallizing.

The basic mode of operation of the pump is as follows: see the figure shown.

The silicone oil located in the boiler (base) of the pump is heated by external banks of heaters. The oil is vaporized as it is heated and rises within the stack. As it reaches the "cold cap" area, it is forced to change direction and becomes

a "vapour jet." This "vapour jet" is in a direction away from the pump inlet port and it expands as it passes from a high pressure to a low-pressure region of the pump. The vapour jet travels at a high velocity and collides with the molecules of gas from within the pump inlet area. The oil molecules combine with the gas molecules and are forced downwards to the base of the pump.

As the combined molecules are re-heated, they separate and the gas molecules are pumped out of the pump via the foreline pumps. The oil vapour hits the walls of the pump, which are cooled. It condenses and runs down to the base of the pump.

This process is continuous while the heaters are switched on and therefore a constant cycle is maintained.

To improve the efficiency of the pump, most designs use multi-stage jets within the stack. Normally, there are three stages within this stack plus a "cold cap" at the top of the pump.

The "cold cap" is used to stop the "backstreaming" of oil back into the chamber (inlet) port. "Backstreaming" occurs when some of the oil molecules pass out of the stack. Instead of being forced down, they keep on travelling upwards towards the chamber.

The cold cap is usually cooled by cooling water fed from the vacuum water manifold. Please refer to the supplied water schematic for the parameters required from the water supply. The water returning to the manifold from the base of the pump should not exceed 130 degrees F.

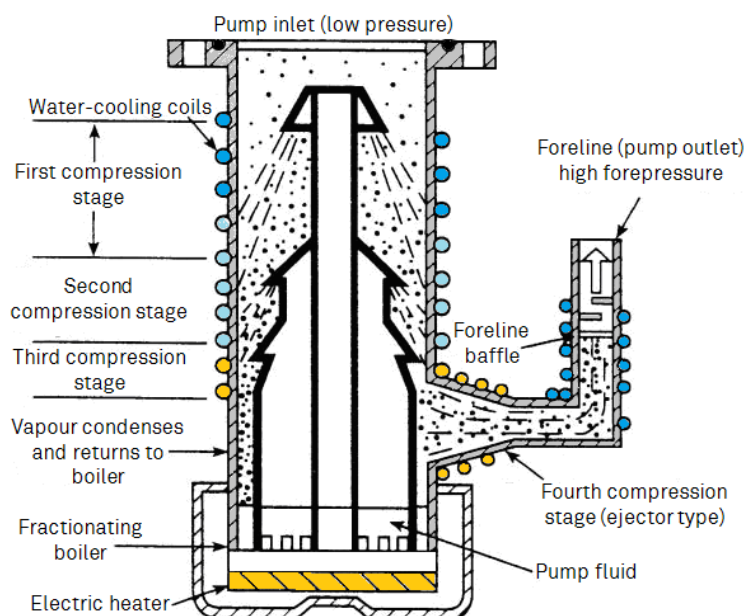


Figure 2.5-3 Oil diffusion pump schematic

Cryogenic Pumping

A minimum of one cryogenic coil system is built within the chamber. This system aids the other pumps to "pump" the large amounts of water vapour that is given off by the substrate.

Within the machine, water moisture can form from the following sources;-

- Released from the substrate being processed.
- From a water leak
- From high humidity air leak

Water will vaporise as the surrounding pressure equals the natural vapour pressure level of water. Water boils at 100 degrees C and the vapour pressure is 1000 mbar, i.e., atmosphere. At 10 degrees C, the vapour pressure is 13 mbar. Therefore, at 13 mbar, the water boils at 10 degrees C and at 10-3 mbar, the water will boil at -78 degrees C.

The normal working pressure in the evaporation zone is below 10-3 mbar and therefore the water will boil off at any temperature above -78 degrees C, unless the chamber can be kept at a temperature of this magnitude, pumping of the chamber becomes a major problem.

Once the water has started to vaporize, it creates an enormous volume change;-

Pressure Level	Volume
$1e^3$ mbar	1245 cc
$1e^2$ mbar	12,450 cc
$1e^1$ mbar	124,500 cc
1 mbar	1,245,000 cc
$1e^{-1}$ mbar	12,450,000 cc
$1e^{-2}$ mbar	124,500,000 cc
$1e^{-3}$ mbar	1,245,000,000 cc

Table 2.5-1 Water volume at various pressure levels

Therefore, if water is being released at a rate of 1 gram/min. at a pressure of 1×10^{-3} mbar, the pumping group will have to pump 1,245,000,000 cc/min. (20,750 litre/sec) of water vapour. This is the approximate capacity of a large booster pump.

To be able to handle the extra pumping requirement, multiple extra pumps would have to be added or a smaller number of cryogenic pumping systems can be used.

The cryogenic pumping system is used to condense moisture on to the coils rather than letting it vaporise and create a large gas load for the pumping system. This is a copper coil that is cooled down to approximately -120 degrees C. At these temperatures, any water released is condensed and frozen onto the coils.

The coil is supplied by a cryogenerator system that is fully interlocked with the rest of the pumping system.

	Note!
	The coils are supplied with a specially super cooled gas from a cryo-generator unit

	Caution !
	Low Temperature Hazard. Do not touch coils when being cooled



Figure 2.5-4 Cryogenic coils in a machine

The cryogenerator unit is controlled from the Machine PLC to switch into the following modes of operation.

Vacuum System Mode	Evaporation Zone Cryogenerator Operation Mode
Vacuum	Cool Mode
Vent	Defrost for three minutes
Vent & chamber open	Standby

Table 2.5-2 Cryogenerator operation table



Figure 2.5-5 Cryogenerator unit

2.6 Vacuum System Valve and Pipework Systems

To allow isolation of the pumping system from the chamber and between sections of the pumping system, isolation valves are used.

Stainless steel pipelines are used to connect the pumps and the chamber.

The current operational state of the vacuum system is displayed on the Vacuum page in the HMI. [See chapter 2.2 Vacuum Pumping System Mimic Display Screen on page 43 for details.](#)

Vacuum Pipeline Isolation Valves

The in-line valves located in the pipe work are activated by air driven actuators.



Figure 2.6-1 Pipeline isolation valve

The pneumatic actuators located either above or on the side of the valves are designed to close the valves in a situation where there is a loss of power to the machine. This is to ensure that the valves return to a safe condition in the case of a major power shutdown situation.

The valves are interlocked by the machine to provide a safe efficient pumping cycle.

High Vacuum Valves (VV7)

This is the plate valve located between the chamber and the diffusion pumps.

The valve above the pump is activated by a mechanical assembly. The mechanism is activated by an external cylinder. This is activated as required by the machine, when the interlocks are in the correct state.

The design of the valves allows them to be either fully opened or fully closed. They will close in the case of a shutdown of the machine's power system and leave them in a safe condition.

There are reed (magnetically activated) switches located at either end of the cylinders to indicate if they are open or closed.



Figure 2.6-2 Plate valves in the closed position

Figure 2.6-3



Figure 2.6-4 Plate valves in the open position

This valve is activated as required by the machine, when the interlocks are in the correct state.

	Warning !
	Do not override the interlocks.

Vent Valves

There are two valves located on the chamber that are used to vent the machine to atmosphere when opened.

Both of these valves have filters attached to stop debris being pulled into the chamber from the surrounding area. It also protects users from trapping their fingers in the valve.

	Warning !
	Do not operate these valves without the filter body in place.

The valves each have open and closed position switches and they are interlocked to the machine control system. If the valves do not close, the chamber cannot be placed into the “Vacuum” mode of operation.

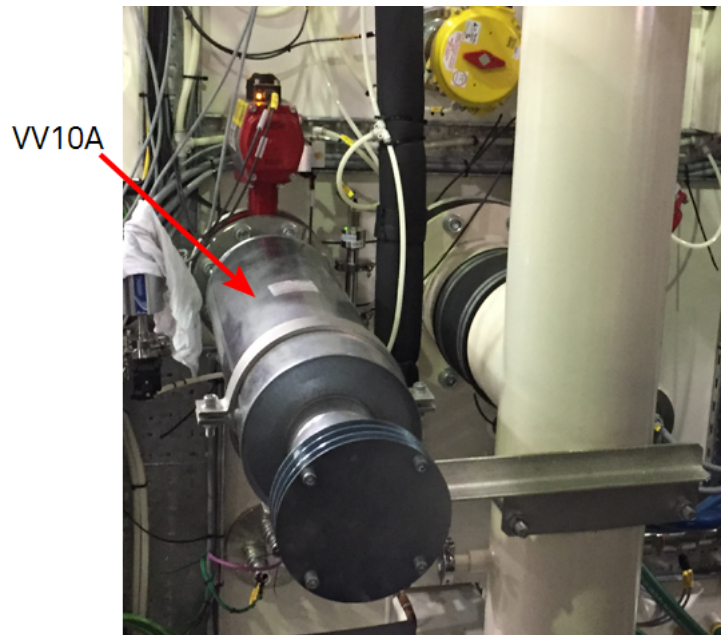


Figure 2.6-5 Photograph of VV10A vent valve

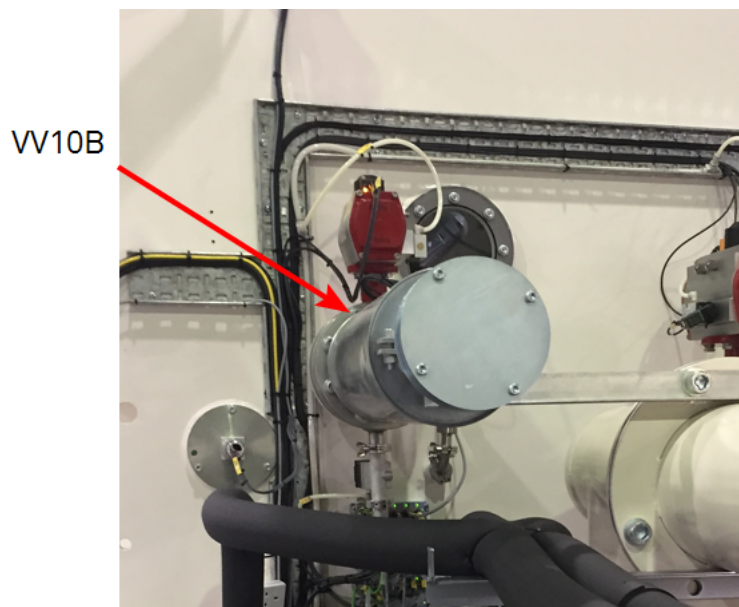


Figure 2.6-6 VV10B vent valve

	Note!
	The winding zone valve (VV10A) will open initially in the vent sequence and then at approximately 400 mbar, the evaporation valve (VV10B) will then open. This mode of operation is normal to minimise the potential of aluminium debris being “blown” around in the evaporation zone of the chamber.

Anti Vibration Bellows

To stop vibration passing between the pumping group and the chamber, flexible bellows are used. These isolate any vibration created by the pumps.

Rubber Bellows

Rubber flexible bellows are used.



Figure 2.6-7 Location of bellows on pumping group

To remove them, loosen the large hose clips; slide the bellows back on to one side of the joint area and remove the pipe adapter from the other side.

The gap in the pipework should be parallel and the maximum gap should be no more than 12mm

Stainless Steel Bellows

To allow for some potential misalignment between the diffusion pumps and the foreline pipework, stainless steel flexible bellows are used.



Figure 2.6-8 Example of the steel bellows in the pumping group

2.7 Vacuum Level Measurement Gauges

Vacuum systems can be described to operate in different types of vacuum environments.

Description	Pressure Range - mbar
Low Vacuum (Rough)	$1e^{-1}$ mbar
Medium Vacuum	$1e^{-3}$ mbar
High Vacuum	$1e^{-3} < e^{-8}$ mbar
Ultra High Vacuum	$< e^{-8}$ mbar

Table 2.7-1 Pressure Range Vacuum Gauges Used

In machines, the coating process takes place in the “high vacuum region” and the vacuum gauges, Bobst Manchester use are shown in the table below.

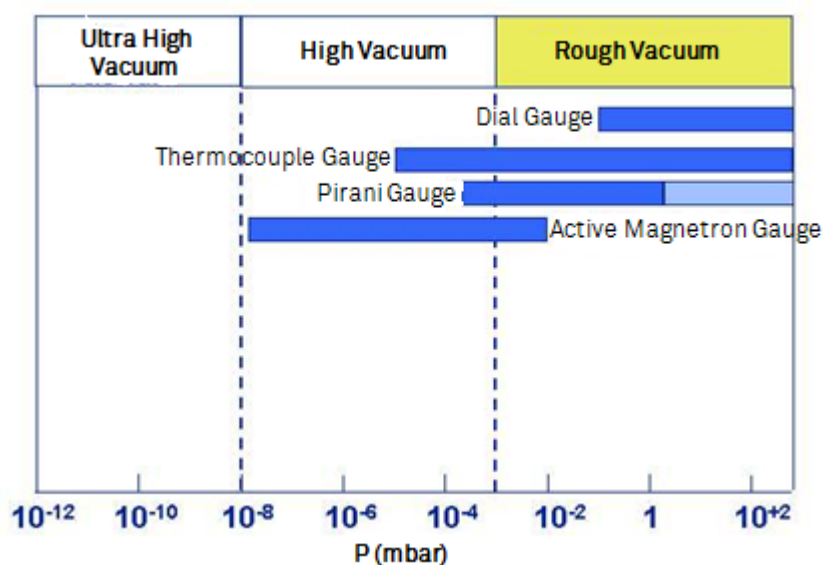


Figure 2.7-1 Approximate vacuum gauge range

Vacuum System Monitoring Overview

The vacuum pressure levels for the machine are displayed on the vacuum screen on the interface screen.

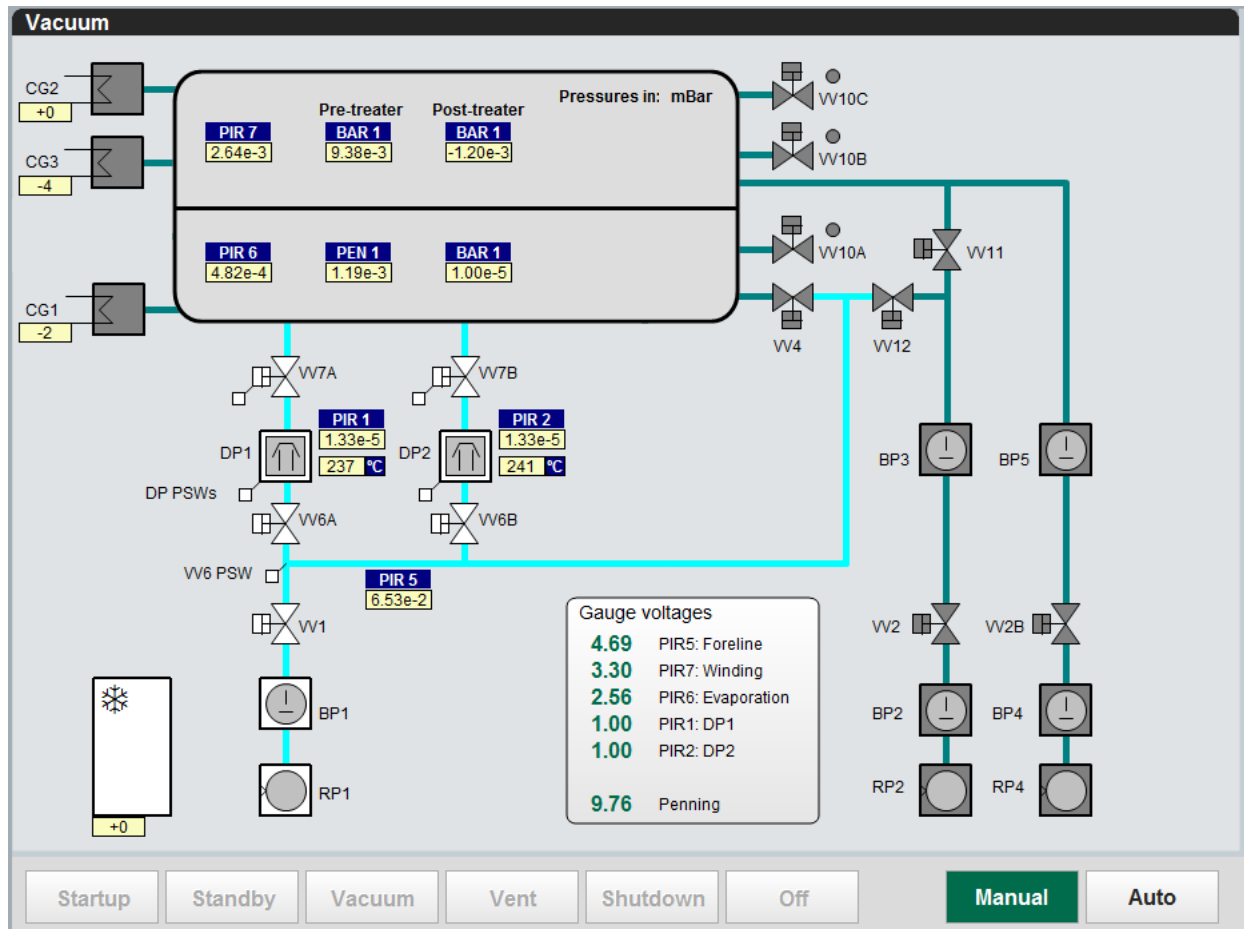


Figure 2.7-2 Vacuum HMI screen showing vacuum levels

A number of different types of gauges are used to monitor the vacuum levels at various points within the vacuum chamber/system. These pressure levels will be indicated in mbar measurements.

Labelling for these various points can be seen on the vacuum HMI screen and can be referred to on the vacuum schematic drawing.

Various pressure level control settings are set to allow automatic features on this machine. These values are compared with the selected vacuum gauge.

The types of gauges that are used in this system are described in the following sections.

MKS Active Pirani Gauges

This type of instrument measures the vacuum level between 200 mbar and $1e^{-3}$ mbar.

They provide information to the monitoring system for interlocking the operation of valves and starting of pumps.



Figure 2.7-3 Schematic of pirani gauge signal monitoring

This type of gauge is located in the following areas of the machine.

- Diffusion pump(s)
- Foreline pipework between the diffusion pump/s and roughing/booster pumps
- Chamber winding zone
- Chamber evaporation zone

	Caution !
	Do not remove the gauges from their location while they are under vacuum.

Dial Gauge

This mechanical type gauge has a range from atmosphere, 1e^{+3} mbar to 1e^0 mbar and is used to monitor the initial evacuation of the chamber.

It is located on the front of the machine above the operator location.

The gauge is provided to allow the user indication of the initial pump down of the chamber.



Figure 2.7-4 Dial gauge indicating the chamber is at atmospheric pressure

This can be used to determine, when to retract the cart from the chamber

	Caution !
	Do not remove the gauges from their location while they are under vacuum.

Active Inverted Magnetron (AIM) Gauge

An Active Inverted Magnetron (AIM) gauge is fitted in the evaporation zone inside the vacuum chamber. It measures vacuum levels in the range below 1 e^{-3} mbar.

The gauge is most commonly used to monitor vacuum levels during the coating process. The pressure is shown on the vacuum screen on the interface screen.

The gauge is not switched on until the VV7 high vacuum valve has opened. When off, its output display on the vacuum scheme will be the same value as the pirani gauge in the evaporation zone.



Figure 2.7-5 Schematic of gauge signal monitoring

Caution !

Do not remove the gauges from their location while they are under vacuum.

Above a certain pressure, the gauge can be damaged, if enabled. Therefore, a plc driven relay is used to enable the gauge when the high vacuum valves open (VV7's) is used.

Gauge Contamination

If oil or process contaminants are able to build up inside a gauge, the insulating properties of the contamination affect the measuring electrodes ability to measure the discharge current.

The only solution for a contaminated gauge is to remove the contamination. The AIM gauge is designed for easy disassembly and cleaning. If cleaning is not a preferred solution, electrode service kits or complete replacement tube assemblies are available.

Baratron Gauge

Baratron gauges are used on machines fitted with a plasma treater. It measures vacuum levels in the range below 1 e^{-3} mbar.

A Baratron gauge is fitted outside the chamber and a tube it passed into the chamber and plasma treater. This is used to measure the vacuum levels within the plasma treater.

The pressure is shown on the DIAGNOSE VACUUM HMI screen.

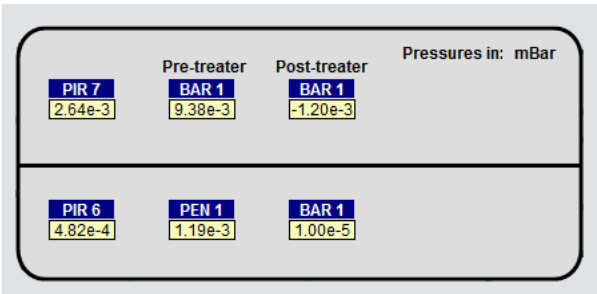


Figure 2.7-6 DIAGNOSE VACUUM HMI screen.



Figure 2.7-7 Baratron gauge

Screens Displaying Vacuum Information



Figure 2.7-8 All screens showing main vacuum pressure levels at the top of the screen

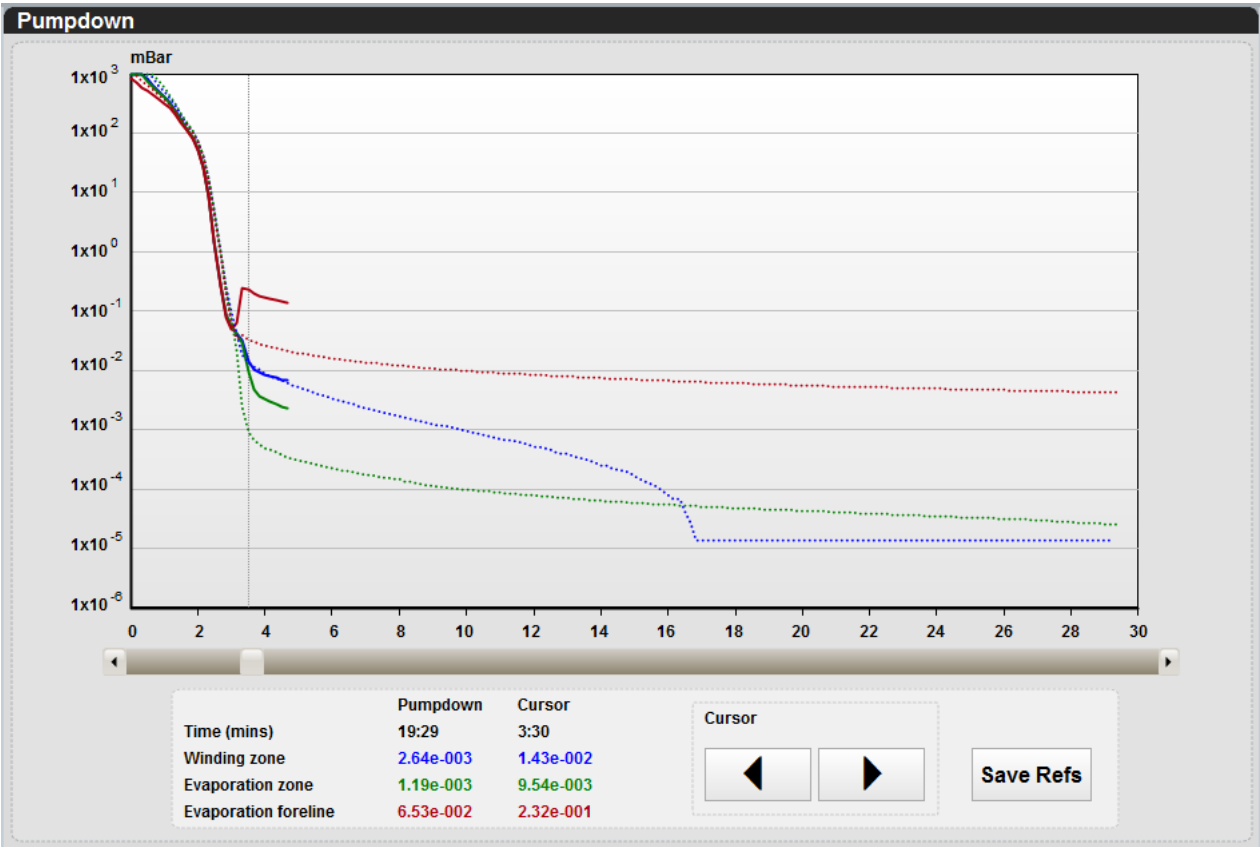


Figure 2.7-9 Pumpdown curve showing vacuum levels

Note!

In the image above, a dotted lines represents reference vacuum levels that were recorded when the machine was clean.

2.8 Chamber Zoning

To provide the most efficient use of the vacuum pumping components, the chamber is split into zones

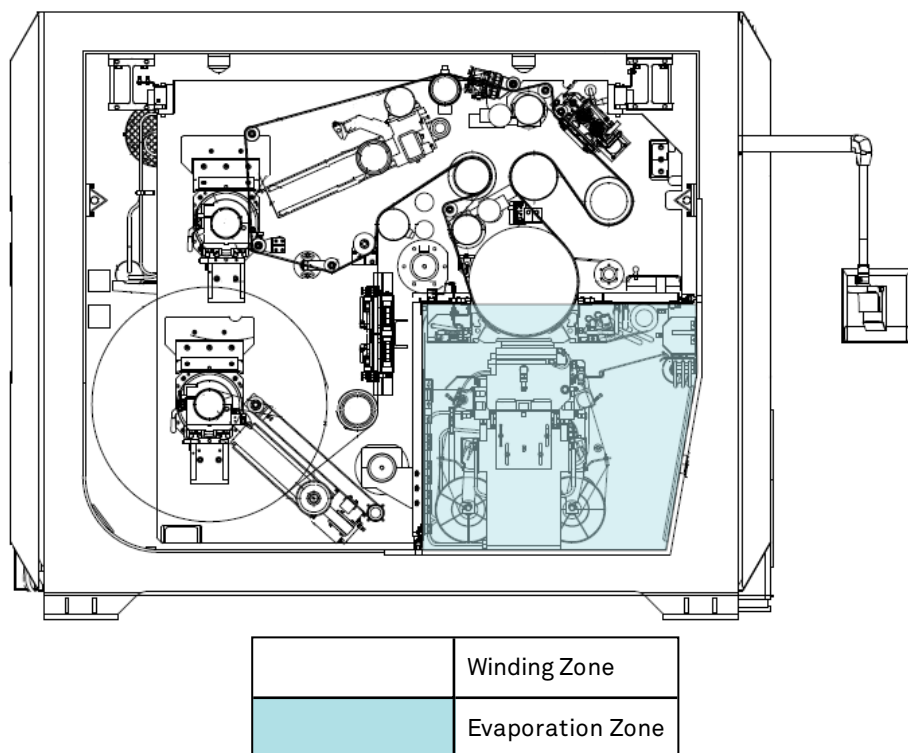


Figure 2.8-1 Cross section of chamber indicating the different zones

A sealing system separates each zone. This sealing system uses strips of flexible shaped rubber arranged along the length of the winding mechanism.



Figure 2.8-2 Flexible seals

Damage to these seals will hinder the effective action of the pumping system.

Other components built into the winding mechanism, such as low conductance seals, around the drum are used to assist in the separation of the zones.

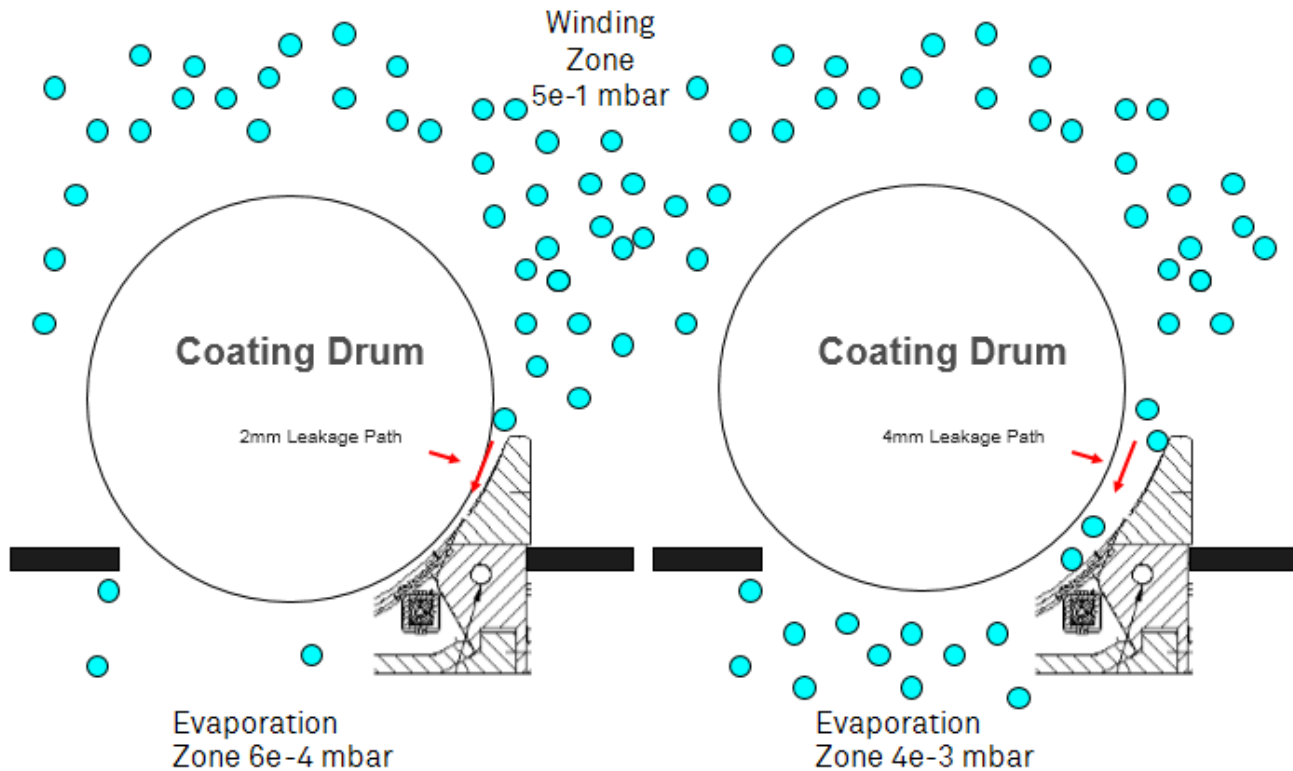


Figure 2.8-3 Effect of coating shield low conductance seals

The assembly around the drum can be lowered to allow cleaning and removal of debris from this area.

The coating shield has to be fully raised to allow the winding cart to be engaged into the machine.

This is achieved by using the RAISE and LOWER push buttons located on the operator interface panel.

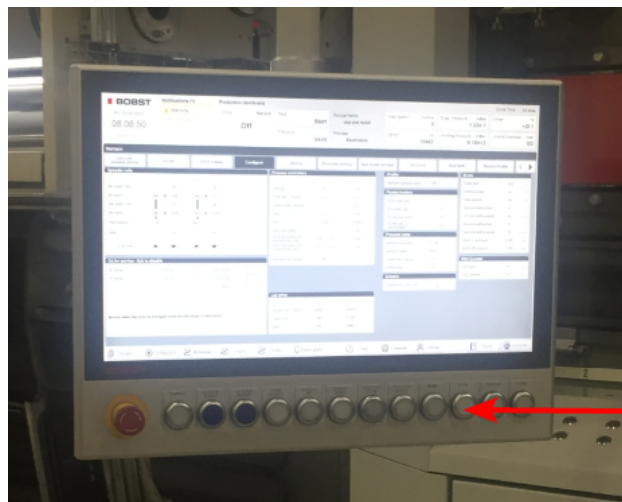


Figure 2.8-4 Location of coating shield raise lower push-buttons

Accessing the Seals

On this machine, the zone sealing arrangement is attached to the drum coating shield. To access the zone sections away from the drum, the coating shield is lowered.

Always check for debris on these shields, as any can cause scratches, as it builds up



Figure 2.8-5 Coating shield in lowered position



Figure 2.8-6 Example of an operator checking the fixed concave shield surface for debris

2.9 Starting the Vacuum Sequence

Under normal process operations, use the following method to start the vacuum sequence after the winding cart has been driven into the chamber:

1. Press the Vacuum and Enable buttons on the operator panel.

This will start the pumpdown sequence and air is removed from the chamber. It will also drive in the winding cart for a few seconds to help form a good seal around the vacuum chamber. If a good seal is not made within 45 seconds then pumpdown will stop automatically.

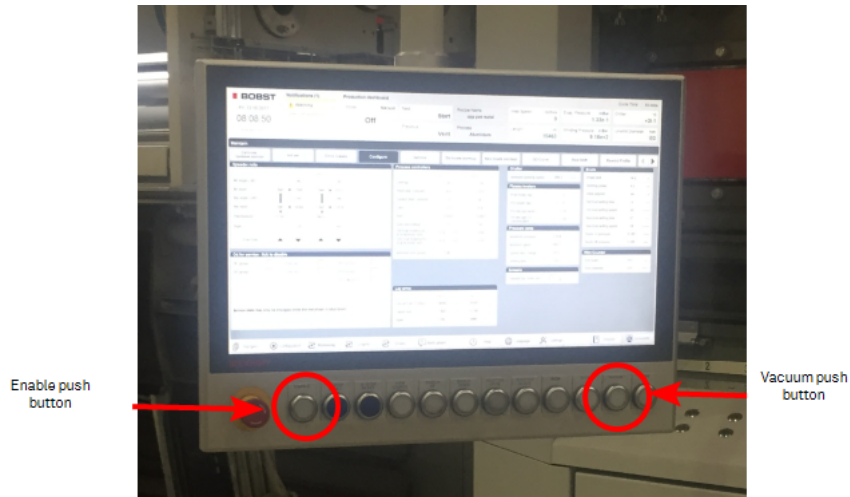


Figure 2.9-1 Operator interface panel

2.10 Pumping System and Chamber Leak Testing Methods

The process is contingent on the vacuum level within the chamber being in the correct vacuum level, i.e. below 2×10^{-3} mbar.

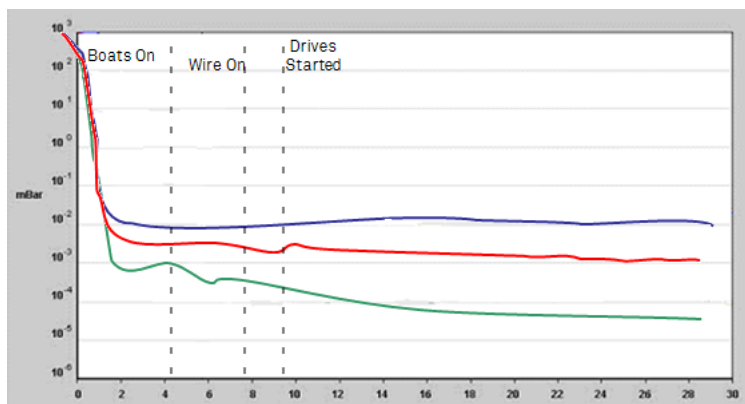


Figure 2.10-1 Curve showing normal pump down cycle for the process

When the required vacuum level cannot be reached, one of the following is normally the reason;

- Air Leak into the chamber
- Internal Water Leak
- Unclean Machine
- Outgassing Substrate
- Pump Failure

Each subject can be explained in a little more detail.

Air Leaks

When there is a suspected leak of air from outside to inside the chamber, the system can be leak checked.

- All Vacuum Fittings-Klein Fittings
- All Vacuum Gauges
- All Rotary Leadthroughs
- All Isolation Valves
- All Electrical Leadthroughs
- All Water Leadthroughs
- All O-ring Joints

Follow the instructions provided with the leak detector's manual. Leak detection using Helium as a search gas should always start at the top of the chamber, as Helium gas rises.

If a leak is found, repair or replace the defective part.

Normally, it can be assumed that welded joints do not often spring leaks. But if one is suspected, it can be checked using welding inspection dye, which is commercially available.